

Garments Manufacturing MIS & Operational Performance Analytics

Project Overview & Key Business Requirements

PROJECT OVERVIEW

A garment manufacturing company operates multiple factories, production lines, departments, suppliers, and buyers. Operational data is maintained across multiple business functions including orders, production, cutting, quality, inventory, purchasing, shipments, employees, attendance, and target planning.

Management currently lacks a centralized analytical view of operational performance.

The objective of this project is to build a data-driven Manufacturing MIS and Operational Performance Analytics system that integrates data from multiple operational sources and provides management with actionable insights into productivity, quality, inventory, workforce utilization, purchasing, and order performance.

BUSINESS OBJECTIVE

The analysis should answer the following key business questions:

1. Production Performance

- What is the total production quantity by factory, line, and date?
- Which production lines are achieving their targets?
- What is the production achievement percentage?
- Which lines consistently underperform?
- Which products/styles generate the highest production volume?
- How does actual production compare with planned production?

2. Efficiency & Productivity

- Which styles have unusually high SMV?
- Where are the major productivity bottlenecks?

3. Quality Performance

- What is the overall defect rate?
- Which lines or products have the highest defect rates?
- What are the most common defect types?
- Which production lines repeatedly generate quality problems?
- What percentage of production is rejected?
- Is there a relationship between production volume and quality issues?

4. Order & Delivery Performance

- How many orders are currently open?
- What is the order quantity versus produced quantity?
- Which orders are at risk of delay?
- Which buyers generate the highest order volume?
- What is the order completion percentage?

5. Inventory Management

- What is the current inventory quantity and value?
- Which materials are overstocked?
- Which materials are below required levels?
- Which materials have high consumption rates?
- Are there slow-moving or obsolete inventory items?
- Which materials require urgent purchasing?

6. Purchasing & Supplier Performance

- Which suppliers provide the highest purchase volume?
- What is the purchase quantity versus received quantity?
- Which suppliers have delayed deliveries?
- Which suppliers have the highest rejection/quality issues?
- What is the supplier fulfillment rate?
- Which suppliers should management monitor closely?

7. Workforce Analysis

- What is the total headcount?
- What percentage of employees are active, resigned, or terminated?
- Which departments have the highest headcount?
- Which factories have workforce shortages?

8. Factory & Line Performance

- Which factory is performing better overall?
- Which production lines are consistently underperforming?
- Which factory has the highest efficiency?
- Which factory has the highest defect rate?
- Which factory has the best production target achievement?
- What are the major operational differences between factories?